# Multi-Prompt Benchmark Report

10 models x 5 prompt styles (code, iambic pentameter, clinical prose, creative writing, math proof) | NVIDIA Jetson Orin Nano 8GB | llama.cpp 0b1bad1 | GUI off | -ngl 99 -fa on --temp 0.3

## 1. Generation Speed (tok/s)

Tokens generated per second during inference. Consistent across prompt types because generation is memory-bandwidth bound (reading weights), not compute bound.

| Model | Params | Quant | Size | Code | Iambic | Prose | Creative | Math | Avg |
|---|---|---|---|---|---|---|---|---|---|
| CodeGemma 2B | 2.51B | Q4_0 | 1.44 GiB | 30.6 | 30.7 | 30.7 | 30.6 | 30.7 | 30.7 |
| Granite 3.0 2B | 2.63B | Q4_K_M | 1.49 GiB | 27.1 | 27.4 | 27.4 | 27.4 | 27.2 | 27.3 |
| Gemma 4 E2B | 4.63B | Q4_0 | 2.63 GiB | 26.7 | 26.8 | 26.7 | 26.8 | 26.7 | 26.7 |
| Gemma 2 2B | 2.61B | Q4_0 | 1.51 GiB | 25.3 | 25.1 | 25.2 | 25.2 | 25.2 | 25.2 |
| LFM 2.5 2.6B | 2.70B | Q4_K_M | 1.55 GiB | 25.3 | 25.2 | 25.3 | 25.3 | 25.2 | 25.3 |
| Qwen 2.5 3B | 3.09B | Q4_K_M | 1.79 GiB | 21.5 | 21.4 | 21.5 | 21.5 | 21.5 | 21.5 |
| Hermes 3 3B | 3.21B | Q4_K_M | 1.87 GiB | 20.8 | 20.8 | 20.8 | 20.8 | 20.8 | 20.8 |
| Llama 3.2 3B | 3.21B | Q4_K_M | 1.87 GiB | 20.8 | 20.6 | 20.8 | 20.8 | 20.8 | 20.8 |
| Phi-3 3.8B | 3.82B | Q4_0 | 2.03 GiB | 20.7 | 21.2 | 20.7 | 20.7 | 20.8 | 20.8 |
| SmallThinker 3B | 3.40B | Q8_0 | 3.36 GiB | 19.2 | 19.2 | 19.3 | 19.2 | 19.2 | 19.2 |

Highlight: CodeGemma 2B leads at 30.7 tok/s avg. The 3B class (Qwen, Hermes, Llama, Phi) clusters tightly at 20-22 tok/s. Gemma 4 E2B (MatFormer) and LFM 2.5 bridge the gap at 25-27 tok/s.

## 2. Prompt Eval Speed (tok/s) — Context Ingestion

How fast the model processes the input prompt. Varies by prompt type due to token distribution and attention patterns. Higher = faster time-to-first-token.

| Model | Code | Iambic | Prose | Creative | Math | Avg |
|---|---|---|---|---|---|---|
| CodeGemma 2B | 438 | 438 | 532 | 589 | 413 | 482 |
| Granite 3.0 2B | 250 | 337 | 348 | 265 | 300 | 300 |
| Gemma 4 E2B | 128 | 125 | 141 | 142 | 104 | 128 |
| Gemma 2 2B | 235 | 236 | 263 | 256 | 218 | 242 |
| LFM 2.5 2.6B | 233 | 250 | 282 | 282 | 183 | 246 |
| Qwen 2.5 3B | 380 | 346 | 397 | 421 | 352 | 379 |
| Hermes 3 3B | 295 | 302 | 410 | 362 | 366 | 347 |
| Llama 3.2 3B | 378 | 429 | 465 | 474 | 385 | 426 |
| Phi-3 3.8B | 329 | 303 | 415 | 448 | 299 | 359 |
| SmallThinker 3B | 289 | 367 | 299 | 310 | 269 | 307 |

Highlight: CodeGemma leads at 482 tok/s avg. Gemma 4 E2B is slowest at 128 tok/s due to thinking model overhead (processes internal reasoning context). Creative/prose prompts generally process faster than code/math across all models.

## 3. Tokens Generated — Output Completeness

Word count of generated output. Lower counts may indicate truncated or incomplete responses. Higher counts suggest more thorough answers (but not necessarily better quality).

| Model | Code | Iambic | Prose | Creative | Math | Avg |
|---|---|---|---|---|---|---|
| CodeGemma 2B | 185 | 40 | 207 | 283 | 74 | 158 |
| Granite 3.0 2B | 257 | 181 | 127 | 131 | 306 | 200 |
| Gemma 4 E2B | 256 | 130 | 197 | 221 | 193 | 199 |
| Gemma 2 2B | 186 | 61 | 268 | 281 | 256 | 210 |
| LFM 2.5 2.6B | 275 | 168 | 180 | 270 | 200 | 219 |
| Qwen 2.5 3B | 219 | 54 | 255 | 291 | 274 | 219 |
| Hermes 3 3B | 175 | 66 | 262 | 278 | 269 | 210 |
| Llama 3.2 3B | 243 | 47 | 250 | 274 | 238 | 210 |
| Phi-3 3.8B | 141 | 126 | 176 | 235 | 222 | 180 |
| SmallThinker 3B | 49 | 49 | 49 | 49 | 49 | 49 |

Highlight: LFM 2.5 and Qwen 2.5 produce the most output (219 avg). CodeGemma produces the least (158 avg) and failed on iambic pentameter (40 tokens of chat template tokens). Gemma 4 E2B and LFM 2.5 are thinking models that spend tokens on internal reasoning.

## 4. Quality Observations by Prompt Type

| Prompt Type | Best Model | Runner-Up | Key Observation |
|---|---|---|---|
| Code Generation | codegemma:2b | granite3-dense:2b | CodeGemma fastest + correct; Granite good general code |
| Iambic Pentameter | gemma2:2b | granite3-dense:2b | Gemma2 best meter (10-syllable); Granite good form |
| Clinical Prose | qwen2.5:3b | gemma2:2b | Qwen most detailed; Gemma2 best structured; all adequate |
| Creative Writing | qwen2.5:3b | lfm2.5:2.6b | Qwen most vivid; LFM good narrative flow |
| Math Proof | qwen2.5:3b | gemma2:2b | Qwen cleanest proof; Gemma2 proper theorem format |

## 5. Summary Recommendations

| Use Case | Best Model | Gen tok/s | Why |
|---|---|---|---|
| Fastest overall | codegemma:2b | 30.7 | Fastest gen + prompt eval, but code-only quality |
| Best 2B generalist | granite3-dense:2b | 27.3 | Consistent quality, fast, good code+prose |
| All-rounder (3B) | qwen2.5:3b | 21.5 | Consistent quality across all 5 prompt styles |
| Code generation | codegemma:2b | 30.7 | Correct output, code specialist architecture |
| Creative/prose | gemma2:2b | 25.2 | Best poetry meter, strong clinical prose |
| Math/proofs | qwen2.5:3b | 21.5 | Cleanest proof structure, correct reasoning |
| Reasoning/thinking | gemma4 E2B | 26.7 | Thinking model with reasoning tokens |
| Most complete output | lfm2.5:2.6b | 25.3 | Highest avg token count across prompts |

## 6. Output Samples

Representative output previews for quality comparison. Truncated to ~400 chars.

### Iambic Pentameter — gemma2:2b (best)

```
The leaves descend, a fiery, vibrant show,
As summer's warmth gives way to chilling breeze.
The harvest's bounty, now all brown and low,
Whispers of winter's icy, silent ease.

The sun descends, its golden rays now faint,
And shadows grow, as nature's colors fade.
The air grows crisp, a frosty, silent paint,
A blanket of white, upon the land is laid.
```

### Iambic Pentameter — codegemma:2b (failed - chat loop)

```

user

assistant

user

assistant

user

assistant

user

assistant
```

````
user

assistant

user

assistant

user

assistant

user

assistant

<
````

## Math Proof — qwen2.5:3b (best)

````
Theorem: √2 is irrational.

Proof by contradiction:

1. Assume that √2 is a rational number. By definition, a rational number can be expressed as a fraction a/b where a and b are integers with no common factors other than 1 (i.e., the
fraction is in its simplest form) and b is not zero.

2. Therefore, we can write √2 = a/b, where a and b are integers with no common factors other than 1, and b ≠ 0....
````

## Math Proof — gemma2:2b (runner-up)

````
## Theorem: √2 is irrational.

**Proof by Contradiction:**

1. **Assumption:** Assume that √2 is rational.
- This means we can express √2 as a fraction in its simplest form, where the numerator and denominator are integers with no common factors other than 1. Let's represent this fraction
as √2 = a/b, where a and b are integers and b ≠ 0.

2. **Square both sides:** Squaring both sides of the ...
````

## Code — codegemma:2b (specialist)

````
Sure, here's an implementation of the `merge_sort` function in Python:

```python
def merge_sort(lst: list) -> list:
"""
Sorts a list of integers in ascending order using the merge sort algorithm.

Args:
lst (list): The list to be sorted.
````
````

```
Returns:
list: A new list containing the integers in ascending order.

Raises:
TypeError: If `lst` is not a list.
...
```

## Creative — qwen2.5:3b (best)

```
The lighthouse stood like a solitary sentinel in the storm, its spire a dark silhouette against the howling tempest. Inside, the keeper, an old woman with weathered skin and eyes that held stories of solitude, sat by the fire, her hands wrapped around a mug of steaming tea. The room was dim, the only light from the flickering flames and the occasional flash of lightning outside.

The door creaked...
```